

EX NO : 6

REGRESSION MODEL-SLR

DATE : 17-07-26

AIM:

To implement a Simple Linear Regression model using a dataset containing one independent variable and one dependent variable, evaluate its performance, and interpret the results .

PROBLEM STATEMENT:

Develop a Simple Linear Regression model to predict the value of a dependent variable using a single independent variable. Display the regression equation, model parameters, performance metrics, and infer the effectiveness of the model.

TASKS:

1. Load the dataset.
2. Select one independent variable (X) and one dependent variable (Y).
3. Perform train-test split.
4. Build a Simple Linear Regression model.
5. Display the slope and intercept.
6. Predict the output values.
7. Calculate performance metrics.
8. Display the best-fit regression line.
9. Write the regression equation.
10. Draw conclusions from the results.

CODE:

```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# -----
# Load Dataset
# -----
# Download Salary_Data.csv from Kaggle or use your own dataset
df = pd.read_csv("Salary_Data.csv")

print("Dataset")
print(df.head())

# -----
# Select Independent and Dependent Variables
# -----
X = df[['YearsExperience']]
y = df['Salary']

# -----
# Train-Test Split
# -----
```

```

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# -----
# Build Linear Regression Model
# -----
model = LinearRegression()

model.fit(X_train, y_train)

# -----
# Model Parameters
# -----
slope = model.coef_[0]
intercept = model.intercept_

print("\nSlope:", slope)
print("Intercept:", intercept)

# -----
# Prediction
# -----
y_pred = model.predict(X_test)

# -----
# Performance Metrics
# -----
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("\nPerformance Metrics")
print("-----")
print("MAE :", mae)
print("MSE :", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)

# -----
# Best Fitting Line Equation
# -----
print("\nBest Fitting Line Equation")
print(f"Salary = {slope:.2f} * YearsExperience + {intercept:.2f}")

# -----
# Plot Regression Line
# -----
plt.figure(figsize=(8,5))

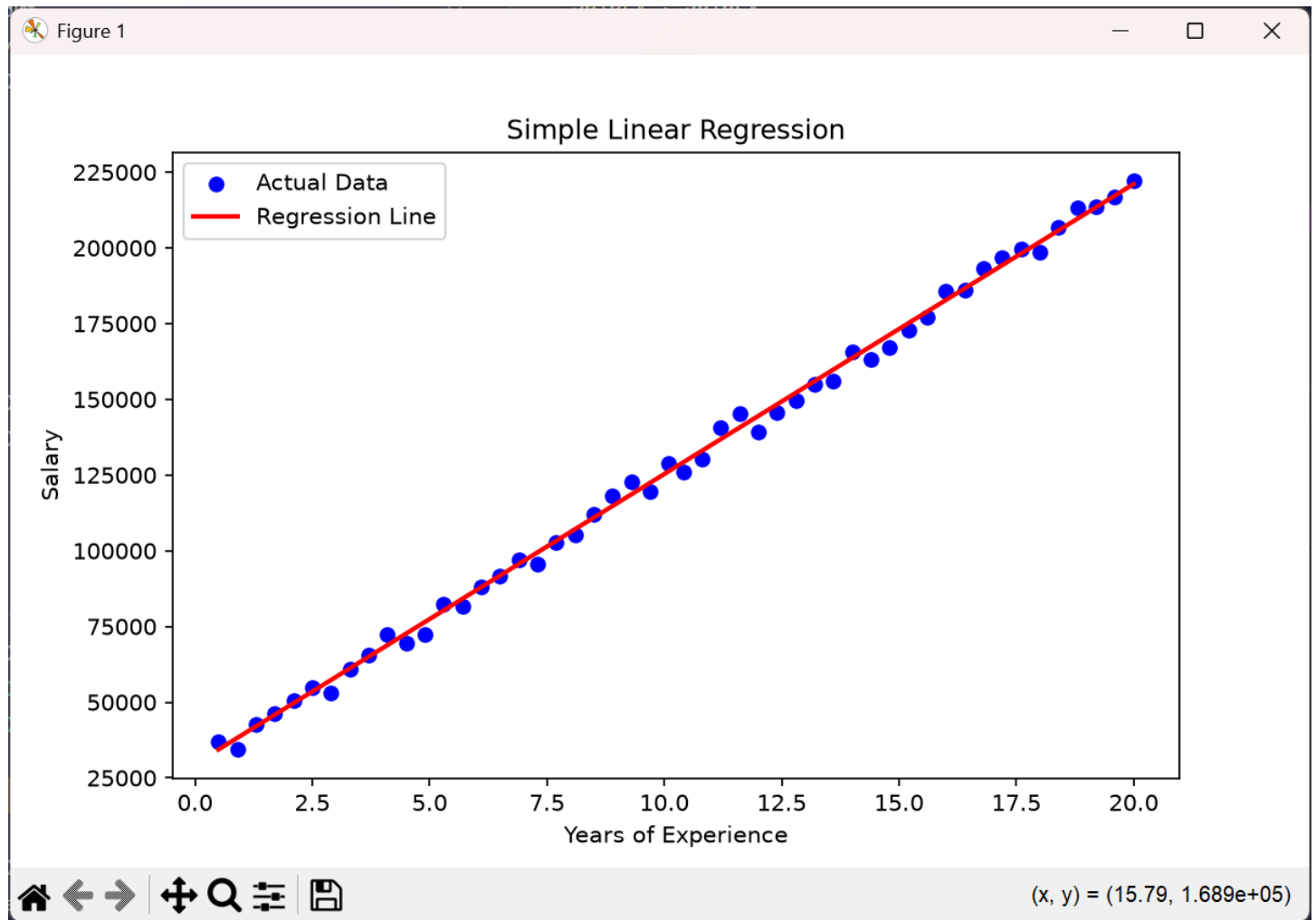
plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X, model.predict(X), color='red', linewidth=2, label='Regression Line')

plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Simple Linear Regression")
plt.legend()

plt.show()

```

OUTPUT:



```
YearsExperience  Salary
0               0.5  37020.0
1               0.9  34410.0
2               1.3  42740.0
3               1.7  46341.0
4               2.1  50684.0

Slope: 9575.608550665607
Intercept: 29582.932502750336

Performance Metrics
-----
MAE : 2271.740905684579
MSE : 6298473.420321912
RMSE: 2509.67595922699
R2 Score: 0.9967459248920739

Best Fitting Line Equation
Salary = 9575.61 * YearsExperience + 29582.93
```